

Operating Systems

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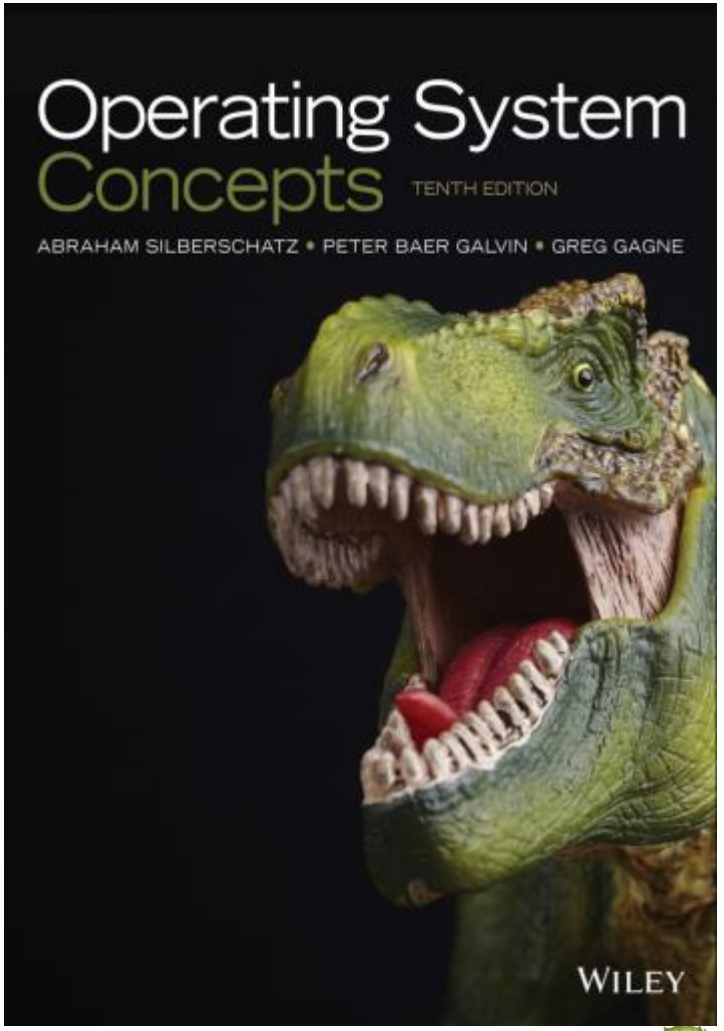
2022





Book

- Operating System Concepts
- Tenth edition
- **Authors:** Peter Baer Galvin, Greg Gagne, Avi Silberschatz







Assessment

- Final : 50
- Attendance & Assignments: 15
- Midterm: 15
- Lab oral : 10
- Lab practical :10





Objectives

- To describe the basic organization of computer systems.
- To describe the services an operating system provides to users, processes, and other systems.
- To discuss the various ways of structuring an operating system.
- To introduce the notion of a process and a thread.



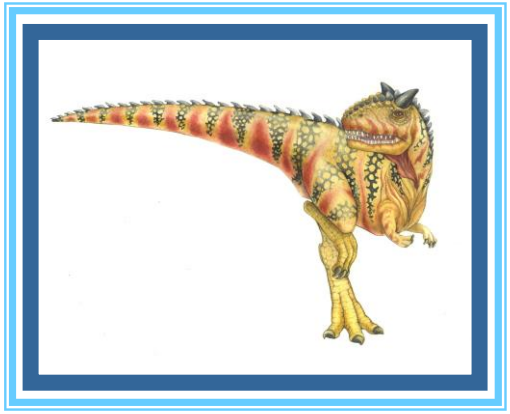


Objectives

- To introduce CPU scheduling, which is the basis for multiprogrammed operating systems.
- To develop a description of deadlocks.
- To provide a detailed description of various ways of organizing memory hardware



Chapter 1: Introduction
Part 1

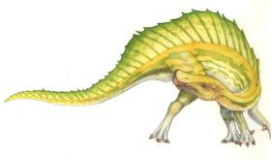


Silberschatz, Galvin and Gagne ©2018



Chapter 1: Introduction

- What Operating Systems Do
- Computer-System Organization
- Computer-System Architecture
- Operating-System Operations
- Resource Management
- Security and Protection
- Virtualization
- Distributed Systems
- Computing Environments





What is an Operating System?

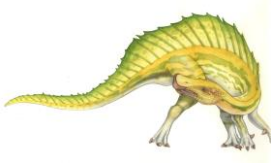
- A program that acts as an intermediary between a user of a computer and the computer hardware
- Operating system goals:
  - Execute user programs and make solving user problems easier
  - Make the computer system convenient to use
  - Use the computer hardware in an efficient manner





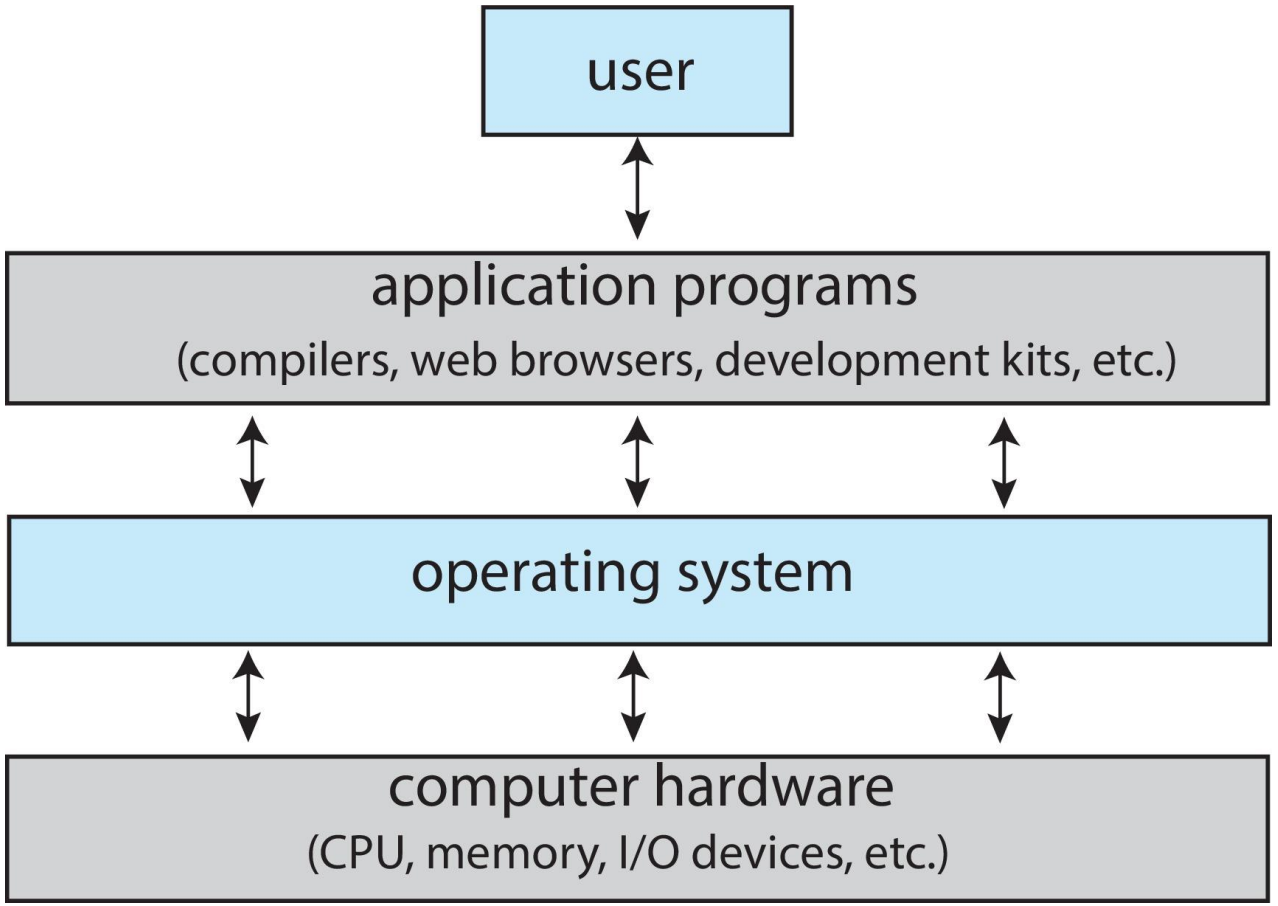
Computer System Structure

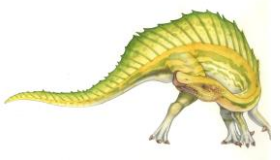
- Computer system can be divided into four components:
  - Hardware – provides basic computing resources
    - CPU, memory, I/O devices
  - Operating system
    - Controls and coordinates use of hardware among various applications and users
  - Application programs – define the ways in which the system resources are used to solve the user computing problems
    - Word processors, compilers, web browsers, database systems
  - Users
    - People, machines, other computers





Abstract View of Components of Computer







What Operating Systems Do

- **User View**
  - Users want convenience, **ease of use** and **good performance**
    - Don't care about **resource utilization**
- **System View**
  - Is the program most intimately involved with the hardware. In this context, we can view an operating system as a **resource allocator**.
  - A computer system has many resources that may be required to solve a problem: CPU time, memory space, file-storage space, I/O devices, and so on. The operating system acts as the **manager** of these resources.





What Operating Systems Do

- OS is a **resource allocator**
  - Manages all resources.
  - Decides between conflicting requests for efficient and fair resource use.
- OS is a **control program**
  - Controls execution of programs to prevent errors and improper use of the computer





Operating System Definition

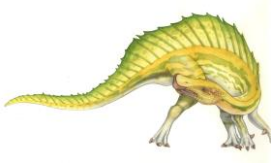
- No universally accepted definition
- "The one program running at all times on the computer" is the **kernel,** part of the operating system
- Everything else is either
  - A **system program** (ships with the operating system, but not part of the kernel) , or
  - An **application program**, all programs not associated with the operating system
- Today's OSes for general purpose and mobile computing also include
  - **middleware** – a set of software frameworks that provide addition services to application developers such as databases, multimedia, graphics





Operating System Definition

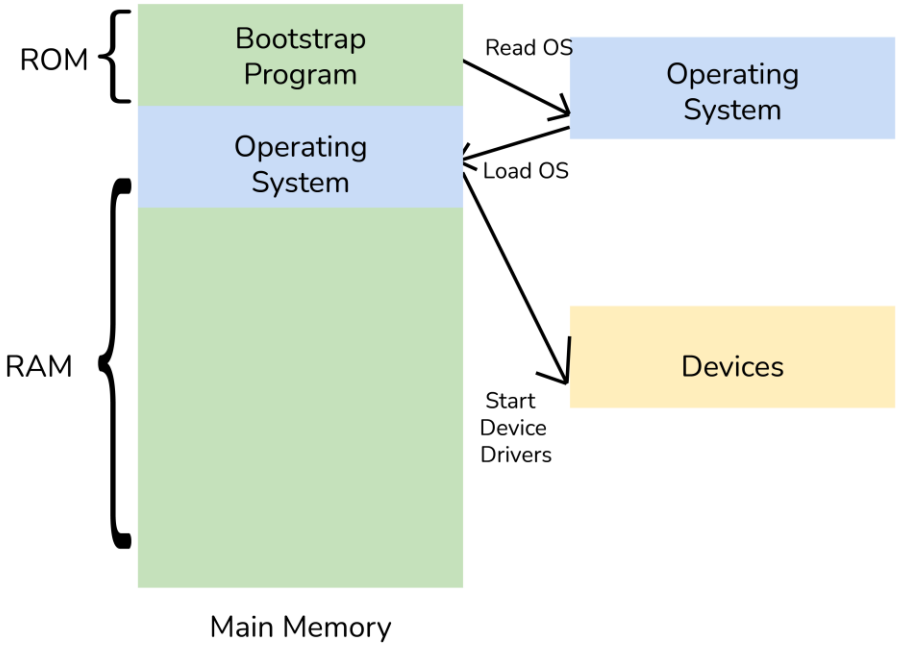
- An operating system is a program that manages a computer's hardware.
  - It also provides a basis for application programs and acts as an intermediary between the computer user and the computer hardware.

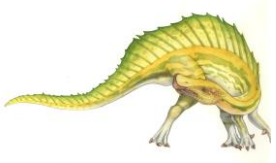




Computer Startup

- **Bootstrap program** is loaded at power-up or reboot
  - Typically stored in ROM or EPROM, generally known as **firmware**
  - Initializes all aspects of system
  - Loads operating system kernel and starts execution







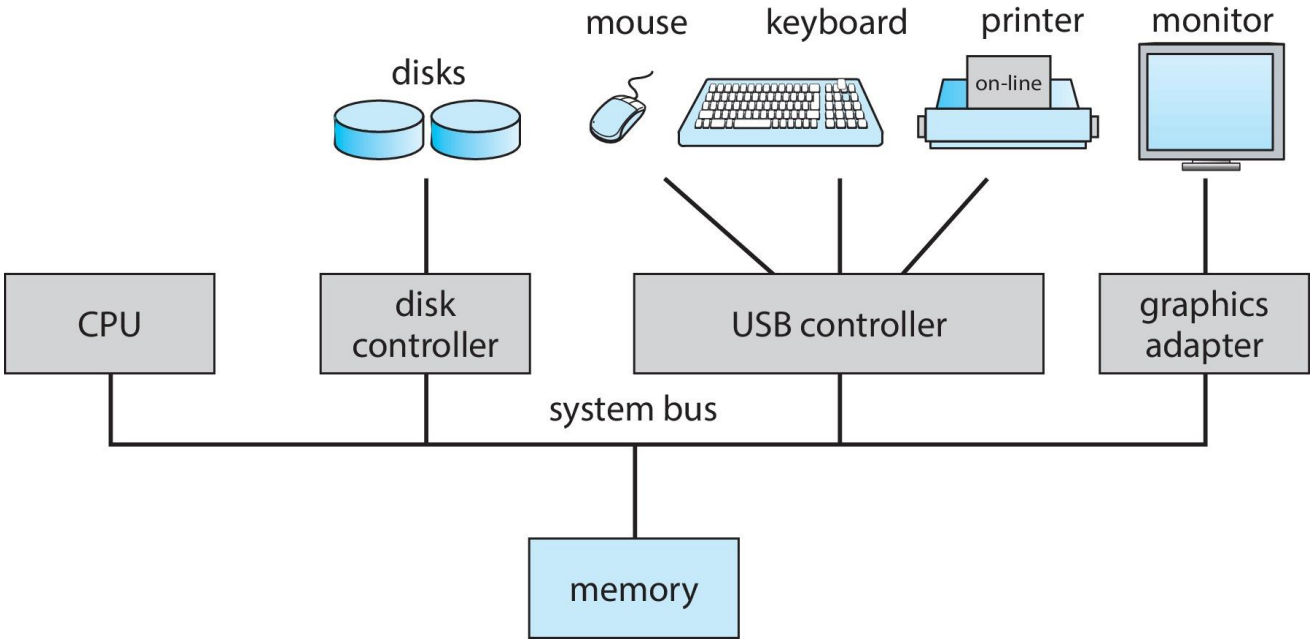
Overview of Computer System Structure





Computer System Organization

- Computer-system operation
  - One or more CPUs, device controllers connect through common **bus** providing access to shared memory
  - Concurrent execution of CPUs and devices competing for memory cycles







Computer-System Operation

- I/O devices and the CPU can execute concurrently
- Each device controller is in charge of a particular device type
- Each device controller has a local buffer
- Each device controller type has an operating system **device driver** to manage it
- CPU moves data from/to main memory to/from local buffers
- I/O is from the device to local buffer of controller
- Device controller informs CPU that it has finished its operation by causing an **interrupt**





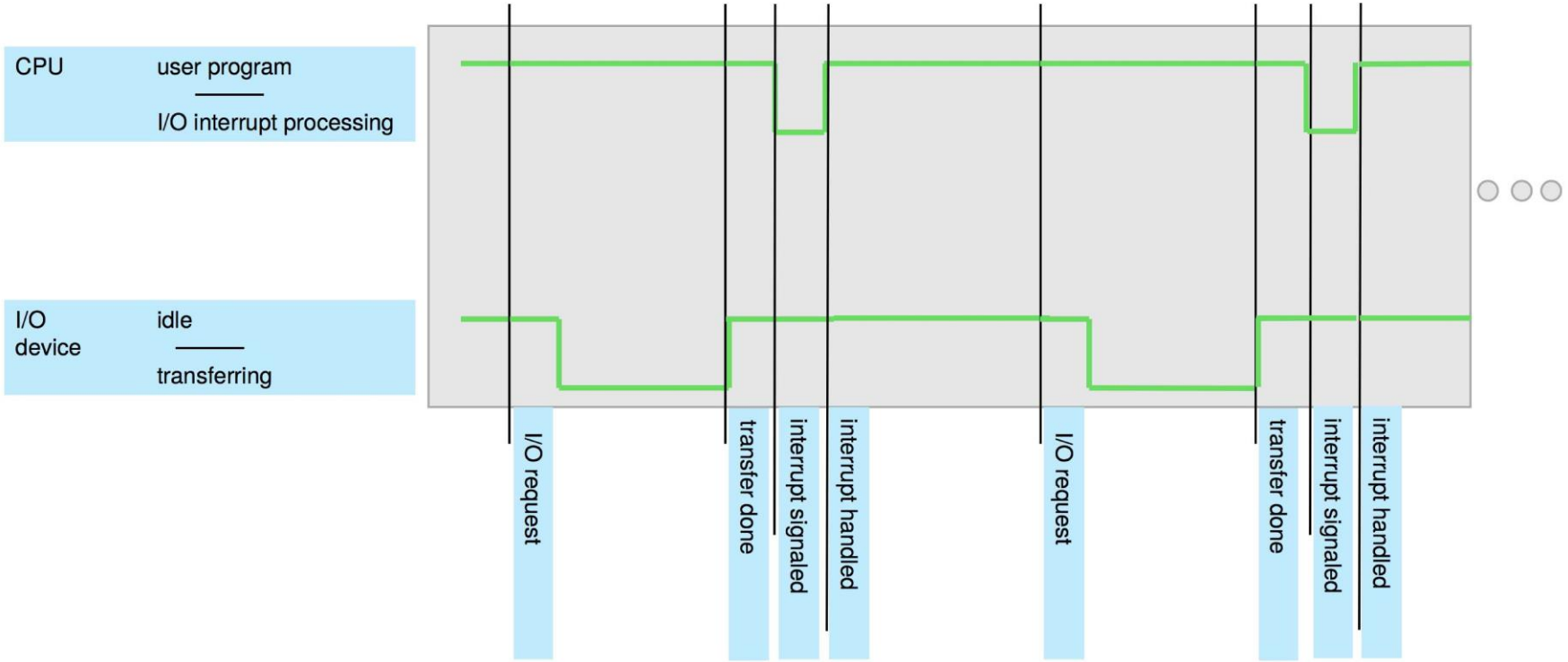
Common Functions of Interrupts

- Interrupt transfers control to the interrupt service routine generally, through the **interrupt vector**, which contains the addresses of all the service routines
- Interrupt architecture must save the address of the interrupted instruction
- A **trap** or **exception** is a software-generated interrupt caused either by an error or a user request
- An operating system is **interrupt driven**
- The operating system preserves the state of the CPU by storing the registers and the program counter





Interrupt Timeline

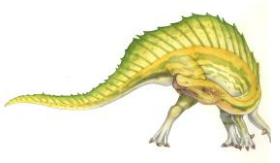






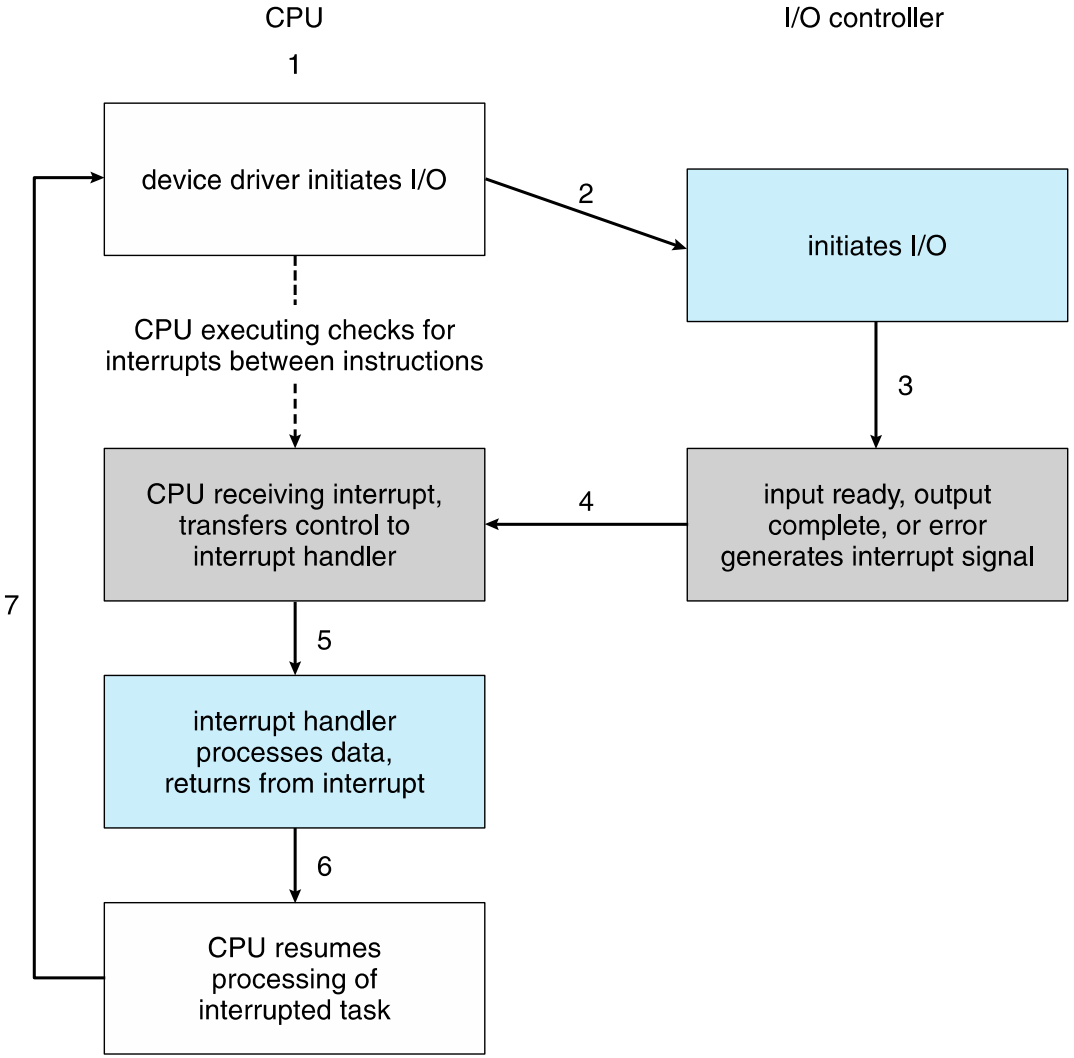
I/O Structure

- Two methods for handling I/O
  - After I/O starts, control returns to user program only upon I/O completion
  - After I/O starts, control returns to user program without waiting for I/O completion





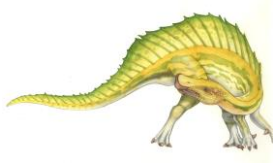
Interrupt-drive I/O Cycle







Storage Structure





Storage Structure

- Main memory – only large storage media that the CPU can access directly
  - **Random access**
  - Typically **volatile**
- **Secondary storage** – extension of main memory that provides large nonvolatile storage capacity





Storage Structure

- **Hard disks** – rigid metal or glass platters covered with magnetic recording material.
  - Disk surface is logically divided into tracks, which are subdivided into sectors.
  - The disk controller determines the logical interaction between the device and the computer.
- **Solid-state disks** – faster than hard disks, nonvolatile.
  - Various technologies.
  - Becoming more popular.

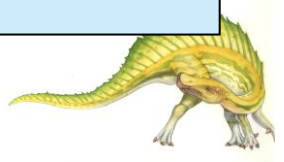




Storage Definitions and Notation Review

The basic unit of computer storage is the **bit** . A bit can contain one of two values, 0 and 1. All other storage in a computer is based on collections of bits. Given enough bits, it is amazing how many things a computer can represent: numbers, letters, images, movies, sounds, documents, and programs, to name a few. A **byte** is 8 bits, and on most computers it is the smallest convenient chunk of storage. For example, most computers don't have an instruction to move a bit but do have one to move a byte. A less common term is **word**, which is a given computer architecture's native unit of data. A word is made up of one or more bytes. For example, a computer that has 64-bit registers and 64-bit memory addressing typically has 64-bit (8-byte) words. A computer executes many operations in its native word size rather than a byte at a time.

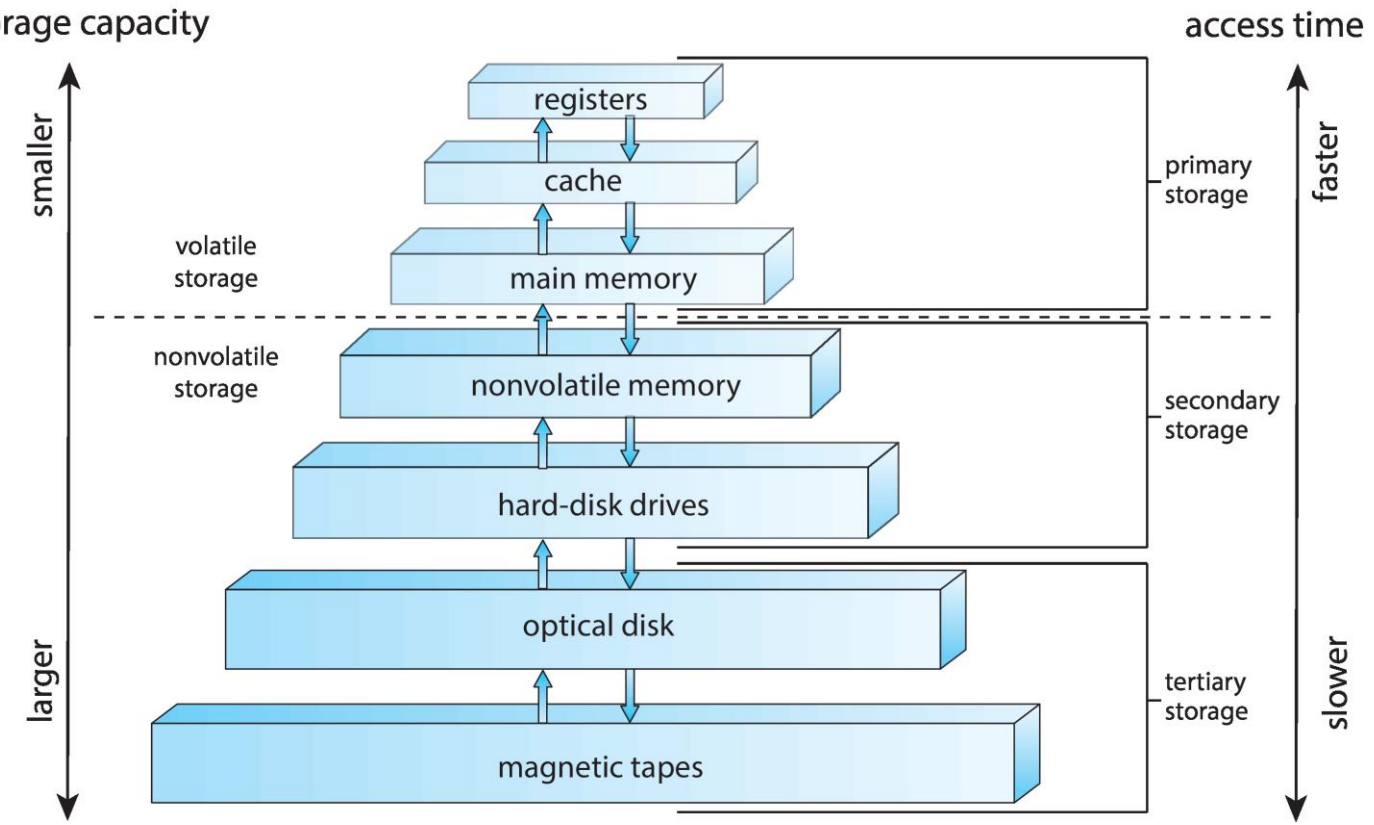
Computer storage, along with most computer throughput, is generally measured and manipulated in bytes and collections of bytes. A **kilobyte** , or KB , is 1,024 bytes; a **megabyte** , or **MB** , is $1,024^2$ bytes; a **gigabyte** , or GB , is $1,024^3$ bytes; a **terabyte** , or **TB** , is $1,024^4$ bytes; and a **petabyte** , or **PB** , is $1,024^5$ bytes. Computer manufacturers often round off these numbers and say that a megabyte is 1 million bytes and a gigabyte is 1 billion bytes. Networking measurements are an exception to this general rule; they are given in bits (because networks move data a bit at a time).





Storage Hierarchy

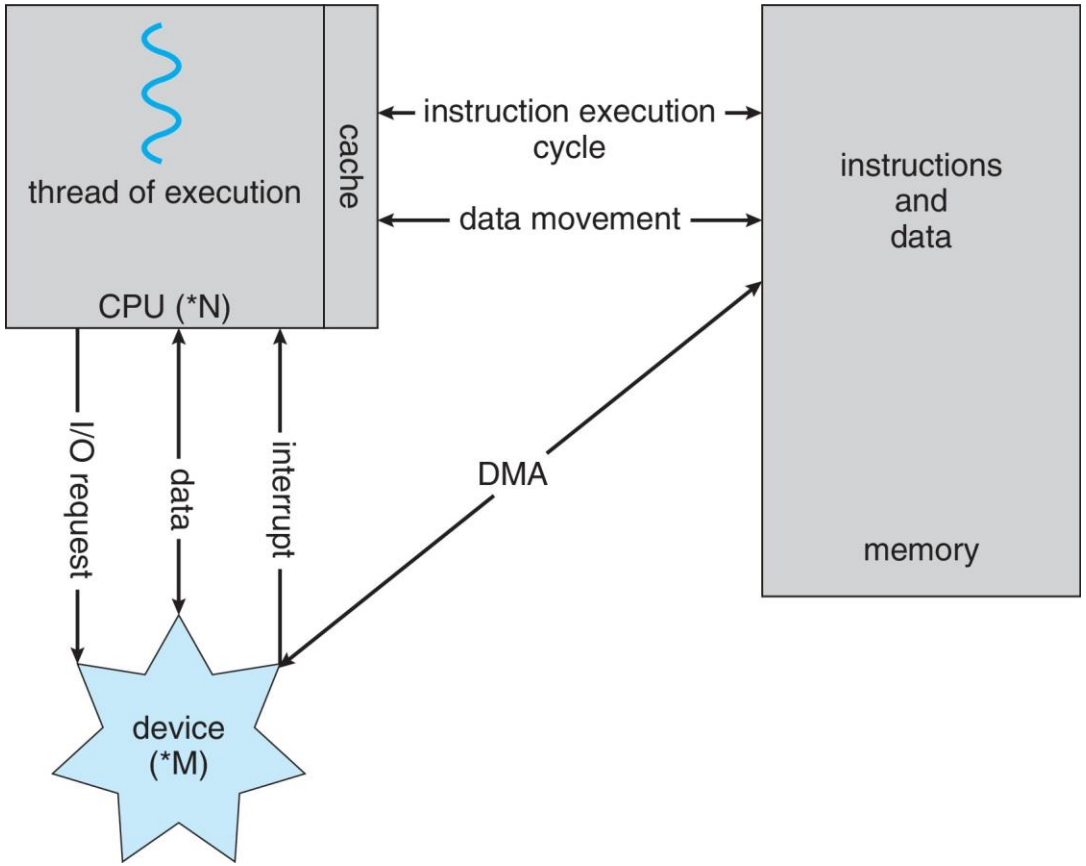
- Storage systems organized in hierarchy.
  - Speed
  - Cost
  - Volatility.







How a Modern Computer Works







Direct Memory Access Structure

- Used for high-speed I/O devices able to transmit information at close to memory speeds
- Device controller transfers blocks of data from buffer storage directly to main memory without CPU intervention
- Only one interrupt is generated per block, rather than the one interrupt per byte





Computer-System Architecture





Single-Processor architecture

- Most computer systems used a single processor containing one CPU with a single processing core.
- The **core** is the component that executes instructions and registers for storing data locally.
- The one main CPU with its core is capable of executing a general-purpose instruction set.





Multiprocessors architecture

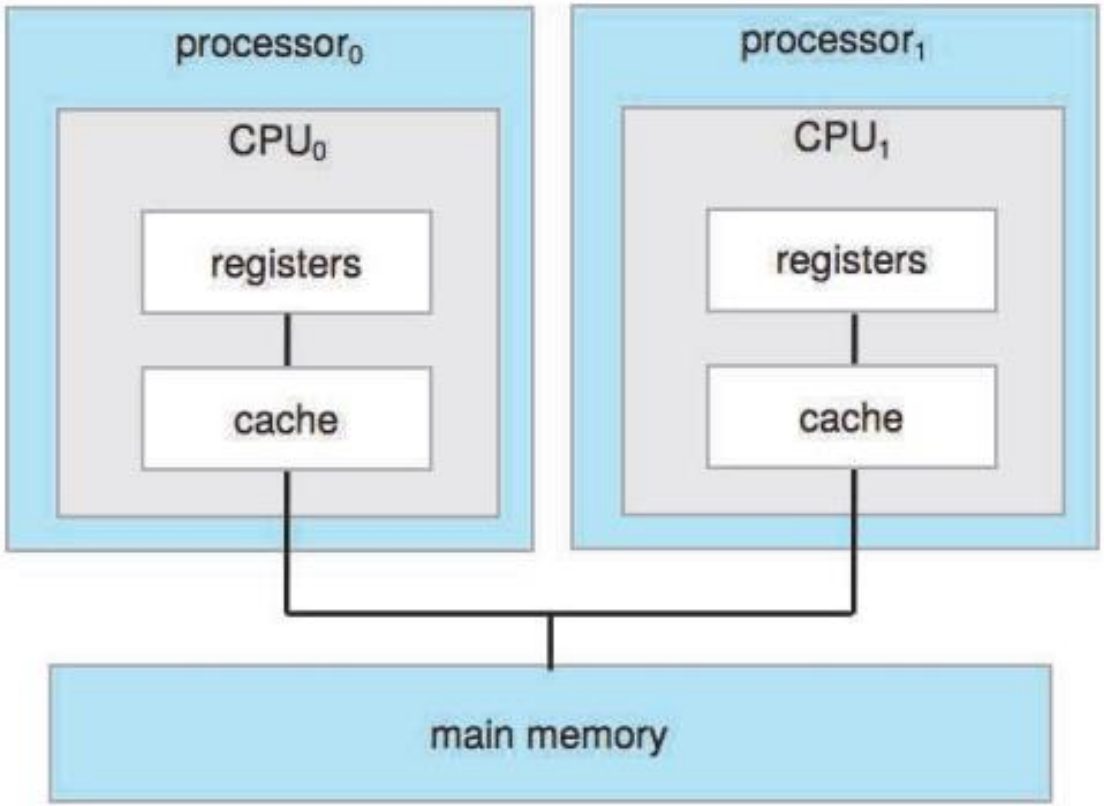
- Multiprocessors systems growing in use and importance
  - Also known as parallel systems.
- Advantages include:
  - **Increased throughput:** by increasing the number of processors, we expect to get more work done in less time.
  - **Economy of scale:** Multiprocessor systems can cost less than equivalent multiple single-processor systems, because they can share peripherals, mass storage, and power supplies.
  - **Increased reliability:** graceful degradation or fault tolerance.





Multiprocessors architecture

- Multiprocessors systems growing in use and importance
  - **Symmetric Multiprocessing** – each processor performs all tasks.

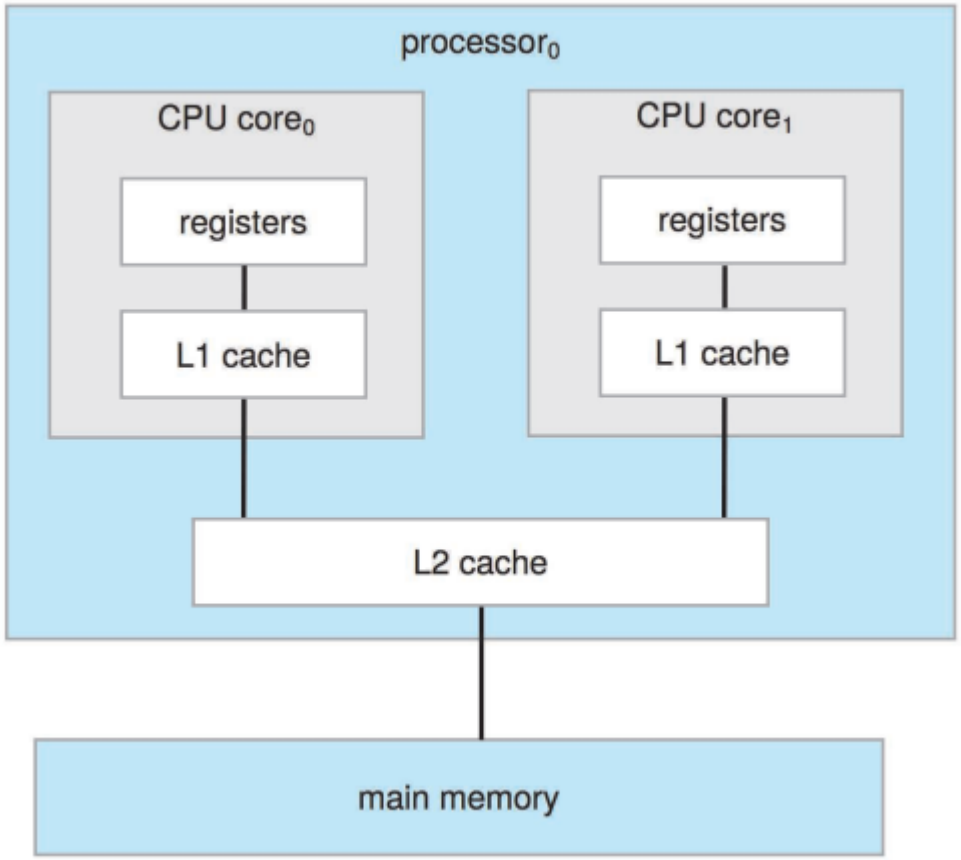






Multicore architecture

- Multiple computing cores reside on a single chip.
  - More efficient than multiple chips with single cores because on-chip communication is faster than between-chip communication

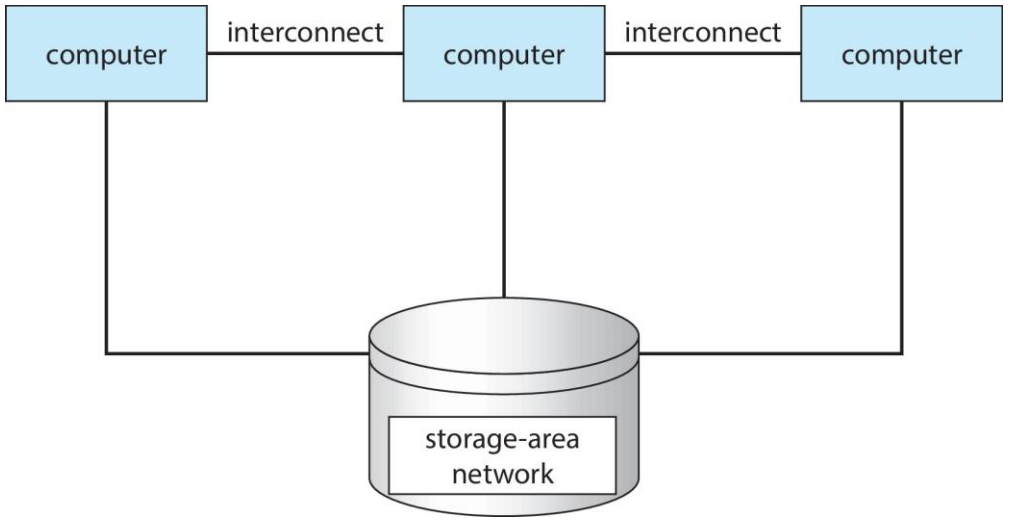






Clustered architecture

- Like multiprocessor systems, but multiple systems working together
  - Usually sharing storage via a **storage-area network (SAN)**
  - Provides a **high-availability** service which survives failures
  - Some clusters are for **high-performance computing (HPC)**
  - Applications must be written to use **parallelization**







THANK YOU

